Notes: Eisfeldt and Papanikolaou (JF, 2013)

Organization Capital and the Cross-Section of Expected Returns

Contents

1 One-sentence summary

This note summarizes Eisfeldt and Papanikolaou (2013), which studies whether organization capital - measured by capitalized SG&A expenditures in Compustat/CRSP firms from 1970 to 2008 - is priced in the cross-section of expected stock returns, develops a production-based model in which key talent and shareholders split rents from organization capital, and finds that firms with high organization capital relative to assets earn substantially higher average returns than low-organization-capital firms.

2 Research question and main answer

Question. Does organization capital create a distinct source of shareholder risk, and does that risk appear in the cross-section of expected returns?

Main answer. Yes. Organization capital is not fully owned by shareholders because it is partly embodied in key employees. When the outside option of key talent improves, shareholders must give up more rents to retain that talent. Therefore, high-organization-capital firms are exposed to a systematic "frontier organization technology" shock, and investors require compensation for bearing this risk.

Empirical headline. A portfolio long firms with high organization capital to assets and short firms with low organization capital to assets earns about 4.7% per year in average returns. The CAPM, Fama-French three-factor model, and Carhart four-factor model do not explain the spread; a two-factor model with the market and the organization-capital factor does much better.

3 Incremental contribution

The paper contributes along four connected dimensions.

1. **New risk channel.** The paper identifies a new source of shareholder risk: systematic changes in the division of surplus between shareholders and key talent. This differs from standard operating leverage or investment-based explanations because the key state variable affects the employee outside option and thus the shareholder share of cash flows.
2. **Asset-pricing model of organization capital.** The paper embeds organization capital into a production-based asset pricing model. Organization capital differs from physical capital because its productivity is firm-specific and embodied in key talent; shareholders do not fully appropriate its rents.
3. **Practical measurement.** The paper constructs a broad firm-level measure of organization capital using capitalized SG&A expenses. This makes the concept measurable for a long panel of public firms.
4. **Joint validation: characteristics, returns, and macro proxies.** The paper does not only show a return spread. It also validates the measure using key-person risk disclosures, manage-

ment quality, IT expenditures, profitability, executive compensation, reallocation, investment in organization capital, and operating leverage patterns.

4 Core intuition

Physical capital is largely owned by shareholders. Organization capital is different: it is embedded in key talent, routines, managerial systems, relationships, training, and firm-specific knowledge. This means that shareholders and employees jointly claim the surplus generated by organization capital.

The crucial shock is the **frontier organization technology shock**, denoted by x_t. When x_t rises, key talent has better outside opportunities. They can leave and deploy their knowledge in a new firm or force the current firm to give them a larger share of the surplus. Therefore, from shareholders' perspective, high-organization-capital firms lose value when the outside option of key talent improves.

This is why the high-minus-low organization capital portfolio, OMK, is expected to be negatively correlated with the frontier shock. If that frontier shock is bad for marginal utility - for example, because restructuring and reallocation are costly in the short run - then high-O/K firms must earn higher expected returns.

5 Model: basic version

5.1 Production technology

A continuum of firms produces a common output good using physical capital K_i and organization capital O_i:

$$y_{i,t} = \theta_t K_i + \theta_t e^{\varepsilon_i} O_i. \tag{1}$$

Here θ_t is aggregate productivity and e^{ε_i} is firm-specific organizational efficiency. Organization capital is productive only when matched with firm-specific key talent.

The aggregate technology shock follows

$$d\theta_t = \sigma_\theta \theta_t dZ_t^\theta. \tag{2}$$

The frontier organization technology shock follows

$$dx_t = \sigma_x dZ_t^x, \tag{3}$$

where x_t captures the efficiency of organization capital if key talent leaves and deploys it in a new firm.

5.2 Stochastic discount factor

The stochastic discount factor is

$$\frac{d\pi_t}{\pi_t} = -r_f dt - \gamma_\theta dZ_t^\theta - \gamma_x dZ_t^x. \tag{4}$$

Here γ_θ is the price of aggregate productivity risk and γ_x is the price of frontier organization technology risk.

5.3 Value of physical capital

Physical capital cash flows are fully owned by shareholders. With risk-adjusted discount rate $r = r_f + \gamma_\theta \sigma_\theta$, the value of physical capital is

$$V_i^K(\theta_t, K_i) = \mathbb{E}_t \int_t^\infty \frac{\pi_s}{\pi_t} \theta_s K_i ds = \frac{\theta_t}{r} K_i. \tag{5}$$

Thus physical capital has exposure to the aggregate productivity shock θ_t, but it does not expose shareholders to the employee outside-option risk x_t.

5.4 Organization capital and the key-talent outside option

The value of organization capital is the value of cash flows while it stays in the firm plus the option value of reallocating to the frontier technology. Key talent can choose the stopping time τ at which to move organization capital to a new firm:

$$V_i^O(\theta_t, O_i, \varepsilon_i, x_t) = \mathbb{E}_t \left[\int_t^\tau \frac{\pi_s}{\pi_t} \theta_s e^{\varepsilon_i} O_i ds + \frac{\pi_\tau}{\pi_t} V^O(\theta_\tau, O_i, x_\tau) \right]. \tag{6}$$

If key talent exercises the outside option, the organization capital operates at frontier efficiency x_τ:

$$V^O(\theta_t, O_i, x_t) = \frac{\theta_t}{r} e^{x_t} O_i. \tag{7}$$

The reallocation threshold solves the value-matching condition

$$V^O(\theta_t, O_i, \varepsilon^*(x_t), x_t) = V^O(\theta_t, O_i, x_t). \tag{8}$$

The paper derives a closed-form solution for the value of organization capital and the reallocation threshold. The key economic content is that the value of organization capital rises with firm-specific efficiency ε_i and with the frontier technology x_t, because x_t makes the reallocation option more valuable.

5.5 Shareholder value of organization capital

Shareholders do not own the full value of organization capital. They own the residual value after compensating key talent for the outside option. Thus shareholder value from organization capital is the difference between total organization-capital value and key talent's outside option:

$$V_i^{OS} = V_i^O - V_i^{\text{outside}}. \tag{9}$$

The paper shows that this shareholder component is decreasing in x_t. When frontier organization technology improves, key talent's outside option rises, and shareholders must concede more rents.

Total shareholder value is

$$V_i^S = V_i^K + V_i^{OS}. \tag{10}$$

5.6 Return exposure and risk premium

Applying Ito's lemma to shareholder value gives a return with two sources of systematic risk:

$$dR_{i,t} = \cdots + \sigma_\theta dZ_t^\theta + \beta_i^x \sigma_x dZ_t^x. \tag{11}$$

All firms have similar exposure to θ_t, but firms differ in exposure to x_t. The loading β_i^x is more negative for firms with higher organization capital relative to physical capital:

$$\frac{\partial \beta_i^x}{\partial (O_i/K_i)} < 0. \tag{12}$$

Therefore the expected excess return can be written schematically as

$$\mathbb{E}_t[R_{i,t} - r_f] = \text{common productivity-risk premium} + \text{frontier-risk premium depending on } O_i/K_i. \tag{13}$$

If the frontier shock has a negative price of risk, then high-O/K firms have higher expected returns.

6 Extended model

The extended model adds endogenous investment and repeated reallocation.

6.1 Production and firm-specific shocks

Output becomes

$$y_{i,t} = \theta_t e^{u_{i,t}} K_{i,t} + \theta_t e^{\varepsilon_{i,t}} O_{i,t}, \tag{14}$$

where $u_{i,t}$ and $\varepsilon_{i,t}$ are mean-reverting firm-specific productivity shocks to physical and organization capital.

6.2 Capital accumulation

Physical and organization capital evolve according to

$$dK_{i,t} = (i_{i,t}^K - \delta_K)K_{i,t}dt, \tag{15}$$
$$dO_{i,t} = (i_{i,t}^O - \delta_O)O_{i,t}dt. \tag{16}$$

Investment in both assets is subject to convex adjustment costs. The cost of increasing organization capital is

$$C_O(i_O, O; \theta) = \theta \frac{c_O}{\lambda_O} i_O^{\lambda_O} e^{\varepsilon} O. \tag{17}$$

The frontier technology shock is mean reverting:

$$dx_t = -\kappa_x x_t dt + \sigma_x dZ_t^x. \tag{18}$$

Reallocation is costly:

$$C_R(O; \theta) = c_R \theta O. \tag{19}$$

6.3 Shareholder value in the extended model

The shareholder value of the firm is

$$V_i^S = \theta_t \left[q(u_{i,t})K_{i,t} + \left(v(\varepsilon_{i,t}, x_t) - \bar{v}(x_t)\right) O_{i,t} \right]. \tag{20}$$

This equation is the extended-model analogue of the basic intuition: shareholders own the full claim to physical capital, but only the residual claim to organization capital after paying key talent its outside option.

The risk premium is

$$\mathbb{E}_t[R_i - r_f] = \gamma_\theta \sigma_\theta + \gamma_x \sigma_x \cdot \text{exposure}_i^x, \tag{21}$$

where exposure_i^x is increasing in the firm's reliance on organization capital. This term drives the cross-sectional prediction.

7 Measuring organization capital

7.1 Capitalized SG&A method

The empirical measure of organization capital is constructed from SG&A expenditures using a perpetual inventory method:

$$O_{i,t} = (1 - \delta_O)O_{i,t-1} + \frac{\text{SG\&A}_{i,t}}{\text{CPI}_t}. \tag{22}$$

The initial stock is

$$O_{i,0} = \frac{\text{SG\&A}_{i,0}}{g + \delta_O}. \tag{23}$$

The benchmark depreciation rate is $\delta_O = 15\%$, and g is set to the average real growth rate of firm-level SG&A, about 10%.

The key ratio used for sorting is

$$\frac{O_{i,t}}{K_{i,t}}, \tag{24}$$

where K is book assets in the main empirical implementation.

7.2 Why sort within industries?

SG&A accounting practices differ substantially across industries. To avoid measuring only industry-level accounting differences, the paper ranks firms by O/K relative to their industry peers. Firms are grouped into the 17 Fama-French industries, sorted into within-industry quintiles, and then pooled across industries.

8 Empirical sample and portfolio construction

The main sample consists of nonfinancial Compustat firms with December fiscal year-ends and common shares traded on NYSE, AMEX, or NASDAQ. The sample contains 5,917 firms and covers 1970 to 2008.

The paper constructs five value-weighted portfolios based on within-industry organization capital ranks. The main long-short portfolio is

$$\mathrm{OMK}_t = R_t^{\text{High } O/K} - R_t^{\text{Low } O/K}. \tag{25}$$

A high return on OMK means high-organization-capital firms outperform low-organization-capital firms.

9 Validation of the organization-capital measure

9.1 10-K risk disclosures

High-O/K firms are more likely to mention the loss of key personnel as a risk factor in 10-K filings. In a random sample of high and low organization-capital firms, 48% of high-O/K firms list key-person risk, compared with 20% of low-O/K firms.

9.2 Management quality

The organization-capital measure is positively related to the Bloom and Van Reenen management quality score. This supports the interpretation that capitalized SG&A captures management and organizational quality.

9.3 IT expenditures

High-O/K firms spend more on IT relative to book assets. In Table III, high-O/K firms spend 2.10% of book assets on IT, compared with 1.17% for low-O/K firms.

9.4 Productivity

High-O/K firms are more productive. They have higher sales-to-assets and higher Solow residuals, consistent with organization capital being a productive input rather than merely an accounting classification.

10 Main asset-pricing results

10.1 Unconditional returns

The high-minus-low organization capital portfolio earns a large average return. High-O/K firms earn about 4.7% more per year than low-O/K firms. This spread is economically large relative to common equity factor premia.

10.2 CAPM, Fama-French, and Carhart alphas

The return spread is not explained by standard factor models. The high-minus-low portfolio has:

- a CAPM alpha around 5.6% per year;
- a Fama-French three-factor alpha around 5.9% per year;
- a Carhart four-factor alpha around 4.2% per year.

These results imply that the organization-capital premium is not simply a size, value, or momentum anomaly.

10.3 OMK as a factor

A two-factor model with the market and the OMK portfolio can price the cross-section of portfolios sorted on O/K. In the model, OMK captures exposure to the frontier technology shock. Empirically, portfolios sorted by their beta with OMK also display return differences, and the two-factor model substantially reduces pricing errors.

11 Additional predictions and evidence

11.1 Executive compensation

The model predicts that aggregate executive compensation should rise when key talent's outside option improves. Because OMK returns are negatively related to the frontier shock, increases in executive compensation should be associated with lower contemporaneous or lagged OMK returns. The paper finds evidence consistent with this prediction using Frydman and Saks executive compensation data.

11.2 Reallocation and restructuring

When the frontier technology shock is high, reallocation becomes more attractive. The model predicts that measures of reallocation should be negatively correlated with OMK returns. The paper examines capital reallocation, CEO turnover, IPO activity, and management buyouts, and generally finds the predicted relation.

11.3 Investment in organization capital

The model predicts that investment in organization capital rises when the frontier technology shock is high. Using accumulated negative OMK returns as a proxy for the frontier shock, the paper finds that firm investment in organization capital is positively related to this proxy, controlling for Tobin's Q, sales growth, profitability, and firm fixed effects.

11.4 Operating leverage

The model predicts that firms with more organization capital have greater sensitivity of earnings to firm-specific output shocks, but not greater sensitivity to aggregate output shocks. This pattern helps separate the paper's mechanism from a simple operating leverage story. Table X supports this prediction.

12 Interpretation

The paper's economic mechanism can be summarized in three steps:

1. Organization capital raises firm productivity but is tied to key talent.
2. Key talent's outside option varies systematically with frontier organization technology.
3. Shareholders of high-organization-capital firms are short an option on key talent's outside opportunity and therefore require a higher expected return.

This mechanism is especially important for modern firms whose productive assets are increasingly intangible and employee-embodied. The paper therefore connects asset pricing, corporate organization, labor economics, and intangible capital measurement.

13 Critical assessment and limitations

- **Measurement error.** SG&A is a broad accounting category. It contains investment in organization capital, but also other operating expenses. The within-industry sorting mitigates but does not eliminate this issue.

- **Interpretation of the OMK factor.** The paper provides strong evidence that OMK is related to organization capital risk, but the factor may still capture other correlated intangible-capital characteristics.
- **Model abstraction.** The model uses a stylized representation of key talent and outside options. Real-world contracts, retention, governance, and labor market frictions are richer.
- **General-equilibrium channel.** The negative price of frontier technology risk is economically motivated by costly reallocation, but its full general-equilibrium foundation is not fully developed in the main paper.
- **External validity.** The empirical analysis is on U.S. public firms from 1970 to 2008. The premium could vary across countries, private firms, or later periods with different labor-market and intangible-capital dynamics.

14 How this paper is useful for future research

This paper is useful for research on intangible assets, key-person risk, corporate governance, labor-market frictions, and asset pricing. It suggests that intangible capital is not only a valuation issue but also a *cash-flow-rights* and *risk-sharing* issue. Future work can extend the idea by using alternative measures of key talent, employee mobility, patent teams, managerial networks, AI-related organizational capital, or labor-contract data to test whether expected returns reflect systematic changes in bargaining power between firms and key employees.

15 Figures and tables

15.1 Table I. Parameters and Table II. Aggregate and Firm-Specific Moments

Table I

Parameters

This table presents the parameters used in the calibration of our model.

Parameter	Symbol	Value
Technology		
Growth rate of TFP-shock	μ_θ	0.00
Volatility of TFP-shock	σ_θ	0.11
Mean-reversion parameter of frontier shock	κ_x	0.10
Volatility of frontier shock	σ_x	0.20
Mean-reversion parameter of firm *O*-specific shock	κ_ε	0.35
Volatility of firm *O*-specific shock	σ_ε	0.45
Mean-reversion parameter of firm *K*-specific shock	κ_u	0.12
Volatility of firm *K*-specific shock	σ_u	0.25
Investment and Reallocation		
Convexity of adjustment costs for investment in *O*-capital	λ_O	3.20
Proportional adjustment cost for investment in *O*-capital	c_O	625
Depreciation rate of *O*-capital	δ_O	0.15
Proportional reallocation cost of *O*-capital	c_R	1.75
Convexity of adjustment costs for investment in *K*-capital	λ_K	1.80

Table I. Parameters

Table II

Aggregate and Firm-Specific Moments

This table presents the target moments used for calibration. We compare the moments in the data versus moments of simulated data. We report median values across simulations, along with the 5% and 95% percentiles. Moments of dividend growth are from the long sample in Campbell and Cochrane (1999). The rate of capital reallocation is from Eisfeldt and Rampini (2006), defined as the sum of sales of PPE plus the sum of mergers and acquisitions to the sum of PPE. The remaining moments are computed using Compustat data over the 1970 to 2008 period (see the Appendix for variable definitions). The firm-specific moments are time-series averages of the median and interquintile range (IQR).

		Model		
Moment	Data	Median	5%	95%
Aggregate Moments				
Mean of dividend growth	0.025	0.028	−0.012	0.056
Volatility of dividend growth	0.118	0.116	0.084	0.278
Mean of investment rate in *O*-capital	0.211	0.199	0.189	0.217
Volatility of investment rate in *O*-capital	0.015	0.009	0.004	0.020
Autocorrelation of investment rate in *O*-capital	0.817	0.861	0.693	0.943
Mean of capital reallocation rate	0.042	0.051	0.001	0.277
Volatility of capital reallocation rate	0.026	0.088	0.003	0.283
Asset Returns				
Mean excess return of market portfolio	0.064	0.049	0.012	0.083
Volatility of market portfolio return	0.171	0.128	0.105	0.159
Firm Moments				
Tobin's *Q* (median)	1.101	1.107	1.018	1.321
Tobin's *Q* (IQR)	0.848	0.614	0.452	0.835
Organization-to-physical capital (median)	1.079	0.678	0.394	1.678
Organization-to-physical capital (IQR)	1.320	0.770	0.444	1.910
Investment rate in *O*-capital (median)	0.222	0.172	0.158	0.197
Investment rate in *K*-capital (median)	0.111	0.099	0.097	0.108
Cash flows-to-capital (IQR)	0.234	0.145	0.119	0.242
Firm size (log) (IQR)	2.207	1.377	1.236	1.487

Table II. Aggregate and Firm-Specific Moments

15.2 Figure 1. Model Solution

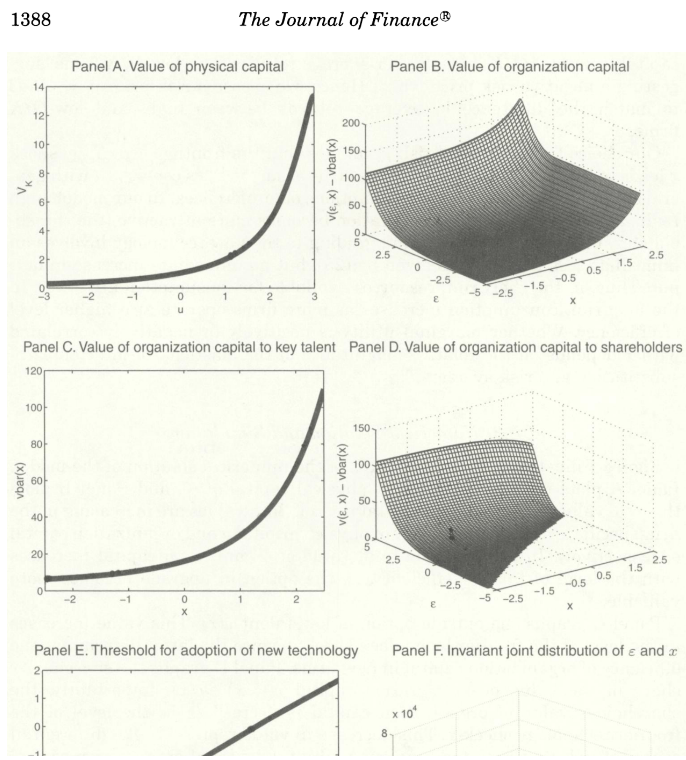

Figure 1. Model solution

15.3 Table III. Firm Characteristics and Organization Capital and Table IV. Asset Pricing: Five Portfolios Sorted on O/K

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Table III
Firm Characteristics and Organization Capital

This table compares characteristics of the five portfolios sorted on organization to physical capital between the data (Panel A) and the model (Panel B). We report the time-series average of the median portfolio characteristic. In the data, we sort firms in five portfolios based on the ratio of organization capital to book assets (Compustat item at), using industry-specific breakpoints. We use the 17 industry classification of Fama and French (1997), and rebalance portfolios every June. In simulated data, we sort firms into five portfolios based on the ratio of organization O_{it} to physical capital K_{it}, and rebalance every year. We report median values across simulations, along with the 5% and 95% percentiles. See the Appendix for variable definitions. The sample period is January 1970 to December 2008.

Portfolio	Low	2	3	4	High
Panel A: Data					
Organization capital to book assets	0.27	0.66	1.09	1.60	2.71
Market capitalization (log, real)	4.89	4.67	4.41	4.10	3.26
Tobin's Q	1.05	1.11	1.18	1.19	1.25
Productivity - sales to book assets (%)	72.38	97.77	111.01	122.99	145.46
Productivity - Solow residual (%)	−11.38	−1.21	2.18	4.02	4.18
Investment to capital (organization, %)	27.11	25.40	22.31	21.35	17.80
Investment to capital (physical, %)	12.63	12.34	12.05	11.60	10.18
Executive compensation to book assets (%)	0.57	0.84	0.89	0.91	1.29
IT expenditures to book assets (%)	1.17	1.69	1.67	1.91	2.10
Labor expense per employee (1,000, real)	54.10	54.60	55.30	56.70	60.10
Capital to labor (log)	3.66	3.28	3.01	2.83	2.56
Physical capital to book assets	38.11	30.72	26.55	24.46	21.30
R&D expenditures to book assets	1.36	2.14	3.17	4.02	6.03
Advertising expenditures to book assets	1.10	1.54	1.88	2.50	3.64
Debt to book assets	29.91	24.69	20.01	17.62	15.07
Panel B: Model					
Organization capital to book assets	0.19	0.42	0.66	1.00	1.65

Table III. Firm Characteristics and Organization Capital

Table IV
Asset Pricing: Five Portfolios Sorted on O/K

ıble shows asset pricing tests for five portfolios sorted on organization capital over assets relative to their industry peers (see notes to Table III), we rebalance portfolios in June every year. Panel A reports results using monthly data, where the sample period is June 1970 to December ?anel B shows results using simulated monthly data from the model, where we report the median coefficient across simulations. In panels 1 we average excess returns over the risk-free rate $E[R] - r_f$, standard deviations σ, and Sharpe ratios SR across portfolios. In panels 2 we report io alphas and betas of a regression of excess portfolio returns on excess returns of the market portfolio. In panels 3 we report portfolio alphas tas of a regression of excess portfolio returns on excess returns of the market portfolio and the OMK portfolio (OMK portfolio corresponds to the ›ortfolio). We include t-statistics in parentheses computed using the Newey-West estimator allowing for one lag of serial correlation in returns. ıualize numbers by multiplying by 12. All portfolio returns correspond to value-weighted returns by firm market capitalization.

	Panel A: Data						Panel B: Model					
io	1	2	3	4	5	5 − 1	1	2	3	4	5	5 − 1
	1. Portfolio moments						1. Portfolio moments					
r_f (%)	4.18	4.54	5.54	5.95	8.81	4.63	4.45	5.47	6.26	7.19	8.69	4.14
	(1.48)	(1.59)	(2.12)	(2.43)	(3.52)	(2.85)	(2.17)	(2.73)	(2.97)	(3.21)	(3.45)	(2.76)
	17.50	17.71	16.26	15.17	15.55	10.10	12.74	12.64	13.19	14.00	15.72	9.67
)	23.89	25.64	34.07	39.22	56.66	45.84	34.93	43.28	47.46	51.36	55.28	42.81
	2. CAPM						2. CAPM					
	−1.19	−0.92	0.57	1.46	4.38	5.57	−0.68	0.42	1.06	1.80	3.02	3.71
	(−1.29)	(−1.07)	(0.67)	(1.54)	(3.69)	(3.47)	(−2.23)	(1.12)	(2.05)	(2.44)	(2.59)	(2.57)
	1.05	1.07	0.97	0.88	0.86	−0.18	0.99	0.98	1.00	1.03	1.08	0.09
	(49.93)	(51.81)	(48.11)	(31.08)	(26.72)	(−4.30)	(40.82)	(34.08)	(25.20)	(17.83)	(11.71)	(0.79)
	90.07	90.90	89.44	83.78	77.62	8.29	97.65	96.63	93.93	88.48	76.74	3.27
	3. Two-factor model						3. Two-factor model					
	0.89	−0.03	−0.32	−0.17	0.89		0.04	-0.10	−0.02	0.05	0.04	
	(1.32)	(−0.04)	(−0.41)	(−0.20)	(1.32)		(0.29)	(−0.29)	(−0.06)	(0.15)	(0.29)	
	0.98	1.04	1.00	0.93	0.98		1.01	0.97	0.98	0.99	1.01	
	(65.09)	(55.79)	(59.22)	(43.02)	(65.09)		(103.29)	(38.63)	(43.06)	(44.83)	(103.29)	
	−0.37	−0.16	0.16	0.29	0.63		−0.19	0.13	0.28	0.47	0.81	
	(−14.83)	(−5.18)	(5.93)	(8.28)	(24.89)		(−11.92)	(3.80)	(9.10)	(15.17)	(50.26)	
	94.33	91.65	90.33	87.23	92.82		99.62	97.58	98.16	98.40	99.76	

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Table IV. Asset Pricing: Five Portfolios Sorted on O/K

15.4 Table V. Asset Pricing: Five Portfolios Sorted on O/K, Controlling for Size, Value, and Momentum and Table VI. Asset Pricing: Five Portfolios Sorted on Beta with OMK Portfolio

Table V

Asset Pricing: Five Portfolios Sorted on O/K, Controlling for Size, Value, and Momentum

This table shows asset pricing tests for five portfolios sorted on organization capital over assets relative to their industry peers (see notes to Table IV for details). In Panel A we report portfolio alphas and betas of a regression of excess portfolio returns on excess returns of the market portfolio and the Fama and French (1993) SMB and HML factors. In Panel B we report portfolio alphas and betas of a regression of excess portfolio returns on excess returns of the market portfolio, the Fama and French (1993) SMB and HML factors, and the Carhart (1997) MOM factor. Data on SMB, HML, and MOM are from Kenneth French's website. The sample period is June 1970 to December 2008. We include t-statistics in parentheses computed using the Newey-West estimator allowing for one lag of serial correlation in returns. We annualize numbers by multiplying by 12. All portfolio returns correspond to value-weighted returns by firm market capitalization.

Portfolio	1	2	3	4	5	5 − 1
			Panel A: Fama-French Three-Factor Model			
$\alpha(\%)$	−1.44	−0.47	0.51	1.92	4.48	5.93
	(−1.60)	(−0.54)	(0.62)	(2.17)	(3.62)	(3.61)
β_{MKT}	1.07	1.05	0.97	0.91	0.89	−0.18
	(50.93)	(49.62)	(43.69)	(42.02)	(25.53)	(−3.99)
β_{smb}	−0.05	−0.03	0.01	−0.23	−0.14	−0.09
	(−1.67)	(−0.80)	(0.16)	(−8.52)	(−3.74)	(−1.53)
β_{hml}	0.05	−0.06	0.01	−0.03	0.01	−0.04
	(1.31)	(−1.80)	(0.15)	(−0.51)	(0.22)	(−0.51)
$R^2(\%)$	90.28	91.02	89.44	86.33	78.59	9.15
			Panel B: Carhart Four-Factor Model			
$\alpha(\%)$	−0.85	0.54	−0.09	1.58	3.33	4.18
	(−0.92)	(0.61)	(−0.10)	(1.76)	(2.63)	(2.55)
β_{MKT}	1.06	1.04	0.98	0.92	0.91	−0.15
	(49.20)	(49.34)	(46.85)	(41.18)	(27.37)	(−3.63)
β_{smb}	−0.05	−0.03	0.01	−0.23	−0.14	−0.08
	(−1.78)	(−1.02)	(0.22)	(−8.22)	(−3.80)	(−1.51)
β_{hml}	0.04	−0.09	0.02	−0.02	0.04	0.00
	(0.98)	(−2.56)	(0.45)	(−0.36)	(0.78)	(0.01)
β_{mom}	−0.05	−0.08	0.05	0.03	0.09	0.14
	(−2.06)	(−3.40)	(1.64)	(0.89)	(2.75)	(3.10)
$R^2(\%)$	90.43	91.43	89.62	86.39	79.29	12.92

portfolios based on their univariate beta with the OMK portfolios also leads

Table V. Asset Pricing: Five Portfolios Sorted on O/K, Controlling for Size, Value, and Momentum

Table VI

Asset Pricing: Five Portfolios Sorted on Beta with OMK Portfolio

This table shows asset pricing tests for five portfolios on their univariate beta with the portfolio of high minus low-O/K firms (see the Appendix for construction details). We compute presorting univariate OMK-betas using weekly returns (see the Appendix for more details) and rebalance portfolios in December every year. In Panel A we report average excess returns over the risk-free rate $E[R] - r_f$, standard deviations σ, and Sharpe ratios SR across portfolios. In Panel B we report portfolio alphas and betas of a regression of excess portfolio returns on excess returns of the market portfolio. In Panel C we report portfolio alphas and betas of a regression of excess portfolio returns on excess returns of the market portfolio and the OMK portfolio. The sample period is January 1971 to December 2008. All portfolio returns correspond to value-weighted returns by firm market capitalization. We include t-statistics in parentheses computed using the Newey-West estimator allowing for one lag of serial correlation in returns. We annualize numbers by multiplying by 12.

Sort	1	2	3	4	5	5 − 1
			Panel A: Portfolio Moments			
$E[R] - r_f$ (%)	3.85	5.42	6.22	6.62	6.76	2.91
	(1.07)	(1.80)	(2.43)	(2.64)	(2.50)	(1.16)
σ (%)	22.34	18.64	15.87	15.53	16.82	15.56
SR	17.23	29.08	39.19	42.63	40.19	18.70
			Panel B: CAPM			
$\alpha(\%)$	−3.08	−0.41	1.13	1.78	1.81	4.89
	(−2.27)	(−0.37)	(1.54)	(1.79)	(1.36)	(2.01)
β_{MKT}	1.31	1.10	0.96	0.92	0.94	−0.38
	(37.19)	(35.02)	(58.80)	(38.96)	(27.29)	(−5.85)
$R^2(\%)$	86.59	87.77	92.56	87.29	77.97	14.58
			Panel C: Two-Factor Model			
$\alpha(\%)$	−0.02	1.47	1.27	0.01	−1.20	−1.18
	(−0.01)	(1.50)	(1.65)	(0.01)	(−1.11)	(−0.64)
β_{MKT}	1.21	1.04	0.96	0.98	1.04	−0.16

Table VI. Asset Pricing: Five Portfolios Sorted on Beta with OMK Portfolio

15.5 Table VII. OMK Portfolio Returns and Executive Compensation and Table VIII. OMK Portfolio Returns and Reallocation

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Table VII

OMK Portfolio Returns and Executive Compensation

This table reports the relation between aggregate executive compensation $\bar{w}$ and the returns of the OMK portfolio, in the data (Panel A) and in the model (Panel B). The OMK portfolio is defined as the portfolio long firms with high organization capital and short (minus) firms with low organization capital to assets. See Section III, notes to Table III, and the Appendix for more details. Data on executive compensation are from Frydman and Saks (2010) and cover the 1970 to 2008 period. Standard errors are computed using the Newey-West estimator allowing for three lags of serial correlation. The last column presents the results of an F-test of whether the sum of the coefficients on the contemporaneous and lagged $-R_t^{OMK}$ are jointly equal to zero.

Compensation to key talent ($\Delta\bar{w}_t$)	$-R_t^{OMK}$	$-R_{t-1}^{OMK}$	R_t^{MKT}	R_{t-1}^{MKT}	$\Delta\bar{w}_{t-1}$	R^2	$p(\mathcal{F})$ OMK=0
Panel A: Data							
Compensation of top three officers, average	−0.172	1.107			0.036	0.353	0.008
	(−0.67)	(4.29)			(0.27)		
	−0.329	1.017	0.189	0.191	0.002	0.428	0.049
	(−1.29)	(3.98)	(1.53)	(1.55)	(0.02)		
Compensation of top three officers, median	0.168	0.418			0.221	0.263	0.004
	(1.11)	(2.82)			(1.63)		
	0.182	0.316	−0.036	0.138	0.230	0.341	0.014
	(1.18)	(2.12)	(−0.50)	(1.95)	(1.78)		
Panel B: Model							
Compensation to key talent, average	1.013	0.986			−0.067	0.403	0.016
	(2.95)	(2.38)			(−0.36)		
	1.113	1.231	0.178	0.261	−0.143	0.484	0.002
	(3.35)	(2.75)	(0.37)	(0.48)	(−0.76)		

executives. Given these two measures of log aggregate executive compensation $\bar{w}_t$, we estimate

$$\Delta\bar{w}_t = a_0 - b_0\, R_t^{OMK} - b_1\, R_{t-1}^{OMK} + c_0 R_t^{MKT} + c_1 R_{t-1}^{MKT} + \rho\Delta\bar{w}_{t-1} + e_t. \quad (38)$$

We estimate the above equation via OLS and report the estimated coefficients, along with Newey-West standard errors, in Panel A of Table IV. We report results with and without controlling for returns to the market portfolio (Table VII)

Table VII. OMK Portfolio Returns and Executive Compensation

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Table VIII

OMK Portfolio Returns and Reallocation

This table reports the relation between measures of capital reallocation and the returns of the OMK portfolio, in the data (Panel A) and in the model (Panel B). See Section III, notes to Tables III and VII, and the Appendix for more details. Data on capital reallocation rate are from Eisfeldt and Rampini (2006) and cover the 1970 to 2008 period; data for CEO turnover are from Execucomp and cover the 1992 to 2008 period; data on initial public offerings are from Ibbotson, Sindelar, and Ritter (1994) and cover the 1975 to 2008 period; data on management buyouts are from Haddad, Loualiche, and Plosser (2011) and cover the 1982 to 2008 period. In Panel A, the first three sets of rows report results using OLS, where the standard errors are computed using the Newey-West estimator allowing for three lags of serial correlation. The last two sets of rows in Panel A report results using a Poisson count regression, with robust standard errors. The last column presents the results of an F-test of whether the sum of the coefficients on the contemporaneous and lagged $-R_t^{OMK}$ are jointly equal to zero.

Reallocation X_t	$-R_t^{OMK}$	$-R_{t-1}^{OMK}$	R_t^{MKT}	R_{t-1}^{MKT}	X_{t-1}	R^2	$p(\mathcal{F})$ OMK=0
Panel A: Data							
Capital reallocation rate	0.002	0.089			0.949	0.832	0.030
	(0.07)	(2.53)			(13.21)		
	−0.001	0.034	0.008	0.022	0.942	0.884	0.088
	(−0.03)	(1.94)	(0.90)	(2.45)	(15.93)		
CEO Turnover	0.009	0.091			0.374	0.462	0.006
	(0.36)	(3.35)			(1.63)		
	0.004	0.14	0.018	−0.034	0.471	0.545	0.012
	(0.12)	(3.35)	(0.87)	(−1.37)	(2.02)		
Number of new initial public offerings (Poisson regression)	2.189	1.267			0.002		0.008
	(2.66)	(1.10)			(4.40)		
	0.911	1.18	1.184	1.188	0.002		0.142
	(0.78)	(1.02)	(1.53)	(1.62)	(3.61)		
Number of new management buyouts (Poisson regression)	1.073	−0.461			0.024		0.042
	(2.87)	(−1.38)			(7.00)		
	1.365	0.793	0.077	−0.942	0.025		0.012
	(2.43)	(1.38)	(0.25)	(−2.68)	(19.66)		
Panel B: Model							
Reallocation frequency	0.214	0.186			0.797	0.737	0.011
	(2.51)	(2.50)			(7.30)		
	0.223	0.176	−0.021	−0.001	0.797	0.777	0.011
	(2.56)	(2.43)	(−0.31)	(−0.11)	(7.31)		
Capital reallocation rate	0.204	0.179			0.785	0.690	0.023

Table VIII. OMK Portfolio Returns and Reallocation

15.6 Table IX. Investment in Organization Capital and Table X. Organizational Capital and Operating Leverage

Table IX

Investment in Organizational Capital

hows estimates of a regression of the investment rate in organizational capital i_{Ot} on lagged values of: minus the accumulated returns of ortfolio $R_t^x \equiv -\sum_{l=1}^{3} R_{t-l}^1 - R_{t-l}^5$, log Tobin's Q, log sales growth $\Delta \log Y_{it}$, and profitability ROA_{it-1}. See Section III and the Appendix for s. Depending on the specification we include firm fixed effects. Panel A shows the empirical results in the data, where the sample period 1970 to December 2008. Panel B shows results in simulated data, where we report the median estimated coefficient and t-statistic across . We winsorize all firm-specific variables at the 1% and 99% level every year. Standard errors are clustered by firm and year.

Panel A: Data					Panel B: Model				
(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
0.177	0.171	0.149	0.177	0.103	0.145	0.158	0.149	0.153	0.144
(2.81)	(3.95)	(2.16)	(2.70)	(2.00)	(4.39)	(4.57)	(4.43)	(4.52)	(4.41)
	0.059			0.034		0.009			−0.008
	(13.04)			(6.30)		(10.00)			(−2.50)
		0.134		0.076			0.013		0.008
		(14.90)		(14.94)			(5.02)		(2.98)
			0.024	0.034				0.033	0.049
			(1.86)	(2.60)				(15.93)	(7.59)
49,523	49,523	49,523	49,523	49,523	52,500	52,500	52,500	52,500	52,500
0.004	0.071	0.113	0.005	0.578	0.137	0.191	0.177	0.241	0.360
–	–	–	–	Y	–	–	–	–	Y

Table IX. Investment in Organization Capital

Table X

Organizational Capital and Operating Leverage

This table compares estimates of operating leverage across the five O/K portfolios in the data and in the model. Panel A shows the estimation results in the data, where the sample period is January 1970 to December 2008. Panel B shows the estimation results in simulated data, where we report the median estimated coefficient and t-statistic across simulations. We compute operating leverage with respect to idiosyncratic shocks as the slope coefficient of a regression of change in log earnings before taxes and depreciation, $\Delta \log E_{it}$, on change in log firm output $Y_{it} = \text{SALES}_{it}$. We compute operating leverage with respect to aggregate shocks as the slope coefficient of a regression of change in log earnings $\Delta \log E_{it}$ on change in log aggregate output or average industry sales $Y_{it} = \{GDP_t, ISALES_t\}$. We interact firm and aggregate output with O/K-quintile dummies, where breakpoints vary by industry. We use the 17-industry classification of Fama and French (1997). In simulated data from the model, firm output y_{it} is given by equation (16); aggregate output $y_t = \int y_{it}\, di$ and earnings are output minus compensation to key talent w. We include firm and O/K-quintile dummies in all specifications. We winsorize all firm-specific variables at the 1% and 99% level every year. Standard errors are clustered by firm and year. See Section III, notes to Table III, and the Appendix for more details.

	Panel A: Data			Panel B: Model	
		Systematic		Firm-Specific	Systematic
$\Delta \log E_{it}$	Firm-specific (Sales)	(GDP)	(ISales)	(y_{it})	(y_t)
$\Delta \log Y_{it}$	1.117	2.954	0.140	1.081	0.963
	(24.84)	(5.28)	(2.44)	(196.91)	(24.24)
$D_2(O/K) \times \Delta \log Y_{it}$	0.108	0.004	−0.024	0.091	−0.005
	(1.99)	(0.01)	(−0.36)	(6.83)	(−0.14)
$D_3(O/K) \times \Delta \log Y_{it}$	0.184	0.001	0.050	0.127	−0.005
	(3.16)	(0.00)	(0.50)	(7.50)	(−0.10)
$D_4(O/K) \times \Delta \log Y_{it}$	0.289	−0.292	0.029	0.178	−0.003
	(5.22)	(−0.66)	(0.34)	(7.85)	(−0.04)
$D_5(O/K) \times \Delta \log Y_{it}$	0.303	0.027	0.032	0.255	−0.001
	(3.82)	(0.06)	(0.41)	(8.16)	(−0.01)
Observations	52,035	52,035	52,035	58,465	58,465
R^2	0.288	0.147	0.141	0.833	0.112

Table X. Organizational Capital and Operating Leverage

15.7 Table XI. Estimating the Market Price of Risk of Frontier Shocks Using Industry Portfolios

Table XI

Estimating the Market Price of Risk of Frontier Shocks Using Industry Portfolios

This table presents GMM estimates of the parameters of the stochastic discount factor $m = a - b_m R_{MKT} - b_x \Delta x$, using the cross-section of 30 industry portfolios (excluding the "Other" industry). We use three sets of proxies for the frontier shock Δx: first, the change in log average (median) executive compensation Δw^a (Δw^m) from Frydman and Saks (2010); second, the change in log total reallocation Δra from Eisfeldt and Rampini (2006); third, minus the returns to the OMK portfolio (see Section III for details). For executive compensation and reallocation we follow the literature on consumption-based asset pricing (see, e.g., Campbell (2003)) and adopt the convention that executive compensation is determined at the beginning of the period, for example, $\Delta x_t = \Delta w_{t+1}$. We normalize a so that $E[m] = 1$ (see, e.g., Cochrane (2001)). We normalize R_{MKT} and Δx to zero mean and unit standard deviation. We report HAC t-statistics computed using errors using the Newey-West procedure adjusted for three lags. As a measure of fit, we report the sum of squared errors (SSQE).

Factor Price	(CAPM)	(1)	(2)	(3)	(4)
R_{MKT}	0.45	0.56	0.67	0.66	0.49
	(4.07)	(4.93)	(5.15)	(4.60)	(4.39)
Δw^a		−0.45			
		(−2.28)			
Δw^m			−0.67		

Table XI. Estimating the Market Price of Risk of Frontier Shocks Using Industry Portfolios